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Using inventory as energy storage for demand-side management of manufacturing operations

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ABSTRACT

Traditionally, electrical grids were designed to have unidirectional flow of power and related services from the utility to the end-user. Modern grids are now tasked with bi-directional flow of power and information. To capitalize on this reality, utilities have designed programs that financially incentivize energy consumers to adjust their energy profile in such a way that is advantageous to the utility. At the same time, users seek to maximize the benefits offered by the utilities. Since this is a complex decision-making process, particularly as users continue to adopt energy storage, solar photovoltaics, and other distributed energy resources, the need for decision support tools is increasingly important. To maximize the financial incentive received by the user, and the benefit gained by the utility, this paper proposes an optimization model to aid manufacturers in operating a battery energy storage device while also using product inventory as additional energy storage. Using inventory as additional energy storage is accomplished by scheduling production to build buffers of inventory during low electricity cost times so that production may be reduced during high electricity cost times. We refer to this problem as the energy job-shop scheduling problem (EJSSP). To accomplish these efforts, a deterministic optimization model is developed in order to determine optimal production and energy storage schedules. It is shown that under the proper conditions, utilities can work directly with energy users such that both parties stand to gain from cooperation with one another. Using several synthetic scenarios, it is found that more flexible manufacturing configurations can have as high as 25% more relative energy cost savings when compared to their more rigid counterparts. Moreover, production scheduling allows inventory to be effectively used as energy storage. Ultimately, the methodology forms the foundation of a gateway for electric utilities and their customers to align their interests and act cooperatively.

1. Introduction

Manufacturing operations naturally seek to generate as much profit as possible at a minimum cost. To minimize costs, there are many decisions that manufacturers can make, such as how to source their materials, manage inventory, and manage their logistics. Another option is to reduce expenses related to the use of energy. The overall costs related to receiving energy is most often, in the case of industrial energy users, the total of two charges: 1) an energy charge for the total amount of energy consumed within the billing period; 2) a demand charge for the largest average power demand within the billing period (typically determined by identifying the largest average amount of energy consumed over any 15-min, sometimes 30-min, interval). Cutting these costs could prove to provide manufacturing firms an advantage over their competitors. Much thought has been given to reducing electrical costs related to heating, ventilation, and air-conditioning (HVAC) as noted in (Abdel-Hadi et al., 2021). For example, (Wang et al., 2021) develop a framework for the control of cooling within a manufacturing environment and (Bhattacharya et al., 2021) design a scheme for the optimal control of a chiller plant that supports a commercial building. One avenue for reducing operating costs that is often overlooked is the electrical expense (demand charges plus energy charges) associated with operating production equipment within manufacturing systems. Straightforward means of capturing these cost savings include energy efficiency measures such as making the switch to higher-efficiency equipment at various points of the manufacturing process. Such maneuvers reduce the overall energy used for the manufacturing and hence help to realize energy cost savings. In addition to these methods, the manufacturer can cash in on opportunities offered by their energy service provider (utility) that allow for a reduced bill in exchange for adjustments to their energy use profile. These methods fall under the

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Abbreviations

$A_{w,s}$	Number of workers type w assigned to shift s
AV_w	Number of workers type w available to work
$BOM_{p,l}$	Number of products p, that product l needs to be made
CA	Battery capacity, in kWh
$CAISO$	California independent system operator
CC_p	The carrying capacity available for product p
CT	Battery charge time
DD	Battery depth of discharge
DER	Distributed energy resource
DP_p	Demand of product p that must be met
DR	Demand response
DRE	Demand response event
DSM	Demand-side management
DT	Battery discharge time
E	Battery efficiency
EI_t	Amount of energy stored at start of time t
$EJSSP$	Energy job-shop scheduling problem
ES	Energy storage
II_p	Initial inventory of product p
$HVAC$	Heating, ventilation, and air-conditioning equipment
$I_{p,t}$	Inventory of product p at the start of time t
IPD	The maximum off-peak power demand
IPD	C Off peak demand charge
$JSSP$	Job-shop scheduling problem
M	A large penalty for deviations from constraints
MA_m	Max number of time periods t that machine m can be on during a day
ND	Number of days in the billing period
$NW_{m,w}$	Number of type w workers needed for machine m to operate
OP_t	Time t is on peak or not
OP	D The maximum on-peak power demand
OPD	C On peak demand charge
PI	Power required to charge
PM_m	Power consumption of machine m when turned on
PO	Power required to discharge
$PR_{m,p}$	Production rate of machine m for product p
RT_t	Cost of energy ($/kWh) during time t
$SH_{s,t}$	Shift s does/does not include time t
TOU	Time-of-use
TT	Amount of time in the time interval t, in hours
W_w	Wage for each different type of worker w
WI_s	Percent wage increase if a worker works schedule s
$X_{m,p,t}$	Machine m is on (1) or off (0) during time t while producing product p

umbrella of demand-side management (DSM).

The aim of DSM is to modify the time pattern and magnitude of participating energy consumers' electrical load (Gelazanskas and Gamage, 2014). The adjustment to the energy consumers' energy use may be tailored to various end goals. Commonly, the main objective is to encourage energy consumers to use less energy during the utility's peak demand times, where the utility observes the largest load to be supplied. Another common goal is to shift energy use to non-peak demand time periods (Nagrath et al., 2008). This helps the utility with reliable operation of the grid, as it helps balance supply and demand during critical periods (Tuballa and Abundo, 2016). There are a few popular methods for attaining this goal.

Demand Response (DR) is a method that has been of growing interest. For example, Lu et al. (2020) details the must-have capabilities of demand response for energy resource aggregators in order to help enable large-scale renewable energy integration. Paterakis et al. (2017) gives a nice general review of demand response. In most DR programs, participants are periodically given signals with information regarding a demand response event (DRE) in exchange for credit towards their electrical bill (Lu et al., 2020). The DRE is characterized by the time the event is to begin, the duration of the event, and the magnitude of load adjustment being requested by the utility (Medha et al., 2019). The DRE for a given day may be set to begin at 11:00, last 30 min, and require a load reduction of 100 kW. These load adjustments can, in some cases, be done directly by the utility, if the energy consumer grants control over some subset of its assets. Direct control, however, is commonly limited to heating/cooling equipment as in Alagoz et al. (2012) and/or the load adjustments are constrained to comfort preferences as in Erdinç et al. (2017). Most facilities enrolled in DR do not participate in direct control, so energy consumers are most often left to their own devices to accommodate the DRE (Medha et al., 2019). This lack of tools available to manufacturers, and others, could be a significant reason why enrollment in DR programs in the industrial sector decreased 29% between 2013 and 2019 by overall capacity, in megawatts (EIA, 2020). The importance of DR is nonetheless expected to grow, as utilities continue to look at it as a resource in their integrated resource planning (Viana et al., 2018). For example, the California Independent System Operator (CAISO) allows DR to be considered as a resource for direct participation in the regional capacity market jointly with other products and the same is true in many other markets across the world (Ribó-Pérez et al., 2021). Hence, it will be of interest to aid users in becoming better equipped for DR participation.

Another form of DSM is the use of time-of-use (TOU) rate plans, where the energy consumer is billed for their electrical usage according to the time of day it is used. In addition to a charge for electricity use, there are separate peak demand charges for the largest average demand for on-peak and off-peak time periods. Energy consumers can thus capture some electrical cost savings for their operation by either rearranging their operation to use more energy during off peak time periods and less during on peak time periods or managing to reduce their peak demands.

In each of the DSM methods mentioned, DR and TOU, the consumer is financially incentivized to participate in, and work towards, the utility's goals, specifically to reduce their demand during certain times of the day that are advantageous to the operations of the utility. Unfortunately, as described in the case of DR, DSM participants are in general left to fend for themselves when it comes to capitalizing on these incentives. This remains the case when considering investment opportunities in distributed energy resources (DERs), such as solar photovoltaics (SPV) and energy storage (ES). In addition to low DSM program participation (Medha et al., 2019), the lack of planning tools can lead to energy consumers making suboptimal financial decisions which also have adverse effects for the utility. The duck curve resultant from the rise of SPV installations is a good example of this.

Models have been developed with the intent of reducing either energy use or energy costs. For example, Mouzon et al. (2007) develop an optimization model for minimizing the electrical consumption of manufacturing equipment while Wang et al. (2019) devise a similar model for machining in production systems. For electrical costs, Rubaiee and Yildirim (2019) developed and solved a multi-objective optimization model for the minimization of energy costs and other attributes in a manufacturing setting while Shrouf et al. (2014) develop a model for the scheduling of a single production machine. Additionally, models have been developed for particularly tasks for DSM efforts. In Javadi et al. (2019) and Javadi et al. (2022), a mathematical model for the optimal sizing and siting of electrical energy storage wile considering time-of-use

pricing is developed. However, the model presented here makes several key contributions. Golmohamadi (2022) provides a thorough review of DSM for manufacturing industries and provides summaries of challenges and opportunities for different methodologies and their goals. While many works make use of optimization for load adjustment, this paper adds to the literature by proposing the use of production scheduling as a way to temporarily defer energy use. Particularly, an optimization model is designed to allow an individual manufacturer to schedule their production along a TOU rate structure with the goal of minimizing their costs. With this capability, the manufacturer can use inventory as energy storage by building up inventory levels during parts of the day where energy costs the least so that energy consumption can be reduced when it costs the most. In turn, this allows the utilities to rely less on peaker-power plants and thus reduces carbon emissions. In addition, energy stored though inventory, the use of a traditional energy storage device (Li-Ion battery) to shift energy is considered. While the importance of considering the stochasticity of a user's load has been shown (Peinado-Guerrero et al., 2021), purely deterministic models are investigated here. The models presented take inspiration from Peinado-Guerrero et al. (2020) where the user is encouraged to use inventory as energy storage in order to decrease load by a given amount for a given amount of time and during a specific time period.

The remainder of the paper is organized as follows: Section 2 give an overview of the methodology used to solve the described problem, Section 3 briefly described the solution methodology, Section 4 presents results on some synthetic cases, and Section 5 provides conclusions and some areas of future research.

2. Overview of the methodology

The goal of this research is to develop an optimization model for the planning of production such that the cost of energy is minimized for a single manufacturer, effectively enabling inventory as energy storage. In other words, the model allows the manufacturer to build up inventory in such a way that energy consumption can be reduced during times of the day where it is most expensive, thereby shifting energy use via storage of product inventory. To make the model robust to various operations, the root of the model is a job-shop scheduling problem (JSSP) formulation. In particular, the model to be presented is lies within a class of optimization problems commonly known as a mixed-integer programming problem. For an overview of integer programming models, see (Laurence A. Wolsey, 2020). While this problem lies at the heart of the formulation, many modifications are made in order to allow for production scheduling along time-of-use price plans. We will refer to this modified problem as the energy job-shop scheduling problem (EJSSP).

Here we detail the EJSSP formulation for the minimization of manufacturer's electrical costs while meeting its production demand. The EJSSP can be described mathematically as follows. There is a set of machines $M = \{1, 2, \ldots, m\}$ in a manufacturing environment. There is a set of products $P = \{1, 2, \ldots, p\}$, each requiring some subset of other products in P, and each product p needs to be processed by some subset of machines in M. Additionally, there is a set of worker types $Q = \{1, 2, \ldots, w\}$ that allows for dictating which pieces of equipment can be operated by different workers. A number of each worker type can be assigned to a shift $S = \{1, 2, \ldots, s\}$, where the shift determines which time periods workers assigned to it will work. While working on a product, each machine m has a certain requirement of workers in W. The planning horizon, T, is discretized into t equal time periods such that $T = \{0, 1, 2, \ldots, t_{end}\}$. Additionally, each machine has a finite buffer capacity where units of product processed by the machine can be stored as they await the next step in their process flow. We assume that if machine m can process multiple products in P, it may process each product at a different speed. However, a variable power consumption is not considered here. There are four sets of decision variables, $X_{m,p,t}$, $I_{p,t}$, $A_{w,s}$, and EI_t. $X_{m,p,t}$ is a binary variable that is one if machine m is working on product p during period t, and zero otherwise. $I_{p,t}$ is the amount of inventory available of product p at the start of period t. OD is the on-peak demand. ID is the off-peak demand. $A_{w,s}$ is an integer variable describing the number of worker type w workers assigned to shift schedule s. EI_t is the amount of energy stored in the Li-Ion battery at time t. Finally, the model has the parameters DP_p, OP_t, $BOM_{p,l}$, ND, etc. as described in the nomenclature. The full mathematical formulation is as follows:

Minimize $z =$

$$ND\left(\sum_{m\in M}\sum_{p\in P}\sum_{t\in T}X_{m,p,t}PM_m RT_t TT + \sum_{t\in T\setminus 0}(EI_t - EI_{t-1})RT_t + \sum_{s\in S}\sum_{w\in Q}A_{w,s}W_w WI_s\right) + OPD\ OPDC + IPD\ IPDC \quad (1)$$

Subject to:

$$I_{p,t_{end}} \geq DP_p \quad \forall p \in P \quad (2)$$

$$I_{p,0} = II_p \quad \forall p \in P \quad (3)$$

$$I_{p,t} = I_{p,t-1} + \sum_{m\in M}X_{m,p,t}PR_{m,p}TT - \sum_{m\in M}\sum_{l\in P\setminus p}X_{m,l,t}PR_{l,p}BOM_{p,l}TT \forall p \in P,\ \forall t > 0 \in T \quad (4)$$

$$\sum_{m\in M}\sum_{l\in P\setminus p}X_{m,l,t}PR_{l,p}BOM_{p,l}TT \leq I_{p,t} \quad \forall p \in P,\ \forall t \in T \quad (5)$$

$$I_{p,t} \leq CC_p \quad \forall p \in P,\ \forall t \in T \quad (6)$$

$$OPD \geq OP_{t-1}\left(\sum_{m\in M}\sum_{p\in P}X_{m,p,t}PM_m + EI_t \Big/ TT - EI_{t-1} \Big/ TT\right) \quad \forall t > 0 \in T \quad (7)$$

$$IPD \geq (1 - OP_{t-1})\left(\sum_{m\in M}\sum_{p\in P}X_{m,p,t}PM_m + EI_t \Big/ TT - EI_{t-1} \Big/ TT\right) \quad \forall t > 0 \in T \quad (8)$$

$$\sum_{p\in P}X_{m,p,0} = 0 \quad \forall p \in P \quad (9)$$

$$\sum_{t\in T}\sum_{p\in P}X_{m,p,t} \leq MA_m \quad \forall m \in M \quad (10)$$

$$\sum_{m\in M}\sum_{p\in P}X_{m,p,t}NW_{m,w} \leq \sum_{s\in S}SH_{s,t}A_{w,s} \quad \forall t > 0 \in T,\ \forall w \in Q \quad (11)$$

$$\sum_{s\in S}A_{w,s} \leq AV_w \quad (12)$$

$$EI_t - EI_{t-1} \leq PI\ TT \quad \forall t > 0 \in T \quad (13)$$

$$EI_t - EI_{t-1} \geq -PI\ TT \quad \forall t > 0 \in T \tag{14}$$

$$\sum_{m \in M} \sum_{p \in P} X_{m,p,t} PM_m TT + EI_t - EI_{t-1} \geq 0 \quad \forall t > 0 \in T \tag{15}$$

$$EI_0 = CA \tag{16}$$

$$EI_{t_{end}} = CA \tag{17}$$

where (1) is the objective function and equates to the electrical and wage related costs over the billing period. In this paper we assume that the investment on energy storage has already been done and it is not part of the problem to be optimized. Constraints 2–6 deal with the flow of inventory through the system. Specifically, constraints (2) ensures that by the end of the daily planning horizon, we have enough products to satisfy the daily demand, constraints (3) sets the initial inventory of each product, constraints (4) is the conservation of inventory constraint, tracking the available inventory for each product from time period to time period, constraints (5) ensures that if any units of a type of product are to be made then enough of its applicable subassemblies must be available in inventory, and constraints (6) ensure that inventory levels always stay below their allotted capacity.

Constraints (7) and (8), along with the objective function, allow for defining the on-peak and off-peak peak demands, respectively. Constraint (9) ensure that no machines are working at time zero. Constraint (10) limit the number of periods that each machine can be on. Constraint (11) state that for each time period the number of workers required by working machines must be limited to the number of workers available. Constraint (12) ensure that the number of assigned workers of each worker type is less or equal than the availability of each worker type. Constraints (13) and (14) deal with the operational constraints of the energy storage unit. Specifically, constraint (13) guarantee the amount of positive change in charge of the unit is limited by its maximum charge capability, and constraint (14) guarantees the amount of negative change in charge of the unit is limit by its maximum discharge capability. Constraint (15) ensure that for each time period the amount of energy consumed by the machines and the amount of energy discharged by the energy storage unit always sum to at least zero. This means that energy is never sent back to the energy service provider. Finally, constraints (16) and (17) ensure that the energy storage unit is charged to capacity at the start of the planning horizon and must be returned to this state by the end of the planning horizon.

The proposed model allows discrete-manufacturing users to minimize their electrical expenses in the face of their production constraints, namely the need to satisfy their demand. In addition to these electrical costs, we consider the cost of incentivizing workers to adjust their work schedule. Particularly, if a worker is assigned to a shift other than their original, a marginal wage increase is considered. Hence, the true objective of the model is to minimize the electrical costs alongside increments in wages due to schedule shifts. To accomplish this, the model must decide how to move products through the system based on demand and worker schedules, and how to utilize the energy storage unit to aid in shaping the electrical profile.

In the next section, the solution methodology for the model is discussed and some parameter values are introduced to develop a test scenario and validate the use of the model.

3. Solution methodology

This section focuses on the discussion of all input parameters and the details of the modeling techniques utilized.

The JSSP is one of the best-known combinatorial optimization problems and is NP-hard, a classification for problems that have no known polynomial time solution procedure (Garey et al., 1976). Even so, optimal solutions for small instances of these problems can be found given the computational capabilities of today's computers. Thus, to solve the mathematical models to optimality, we make use of the standard procedures available in commercial software packages to solve mixed-integer programs.

Pyomo, a Python library used for modeling mathematical programs is used to interface with a commercial solver for the programs at hand (Hart et al., 2017). In particular, Gurobi, an optimization software package, is utilized to solve various problem instances (GUROBI OPTIMIZATION INC., 2014).

In the next session, we run some numerical experiments and discuss the results.

3.1. Input data

To demonstrate the use of the model, we consider three separate synthetic scenarios that are designed to be representative of three different types of generic discrete-manufacturing configurations. Indeed, the goal here is to demonstrate that the model can be used for a wide array of customers, since we exemplify with the most generalized case: manufacturers of products that follow a sequence of generic manufacturing steps. These three scenarios are discussed in more detail in the next section.

Additionally, each scenario takes as input the same rate structure to be billed by for electrical energy use. The billing structure is divided into four components: 1) on peak usage charge, a charge per kilowatt-hour consumed during the on-peak period; 2) off-peak usage charge, a charge per kilowatt-hour consumed during the off-peak period; 3) on-peak demand charge, a charge per kilowatt for the highest average energy use for any period within the on-peak period; 4) off-peak demand charge, a charge per kilowatt for the highest average energy use for any period within the off-peak period. Note that the on-peak and off-peak periods are disjoint subsets of periods of the day.

4. Results

4.1. Exemplary model

To exemplify the model, we first introduce a simplistic 'toy' instance. In this instance, we have three machines, working to produce a single product. The product can take one of two paths before its final step at machine m3. Additionally, each machine has identical processing times and can store inventory once it has been processed. Fig. 1 illustrates this simple generic manufacturing system, where the triangle represents the inventory buffer.

Because the use of traditional energy storage, in addition to inventory energy storage, is of interest to make decisions that adjust the shape of a user's load, once the instance was developed it was solved several times while increasing the amount of energy storage available for use. Fig. 2 shows optimal electrical costs for the billing period with different amount of battery energy storage available for use. Fig. 3 shows the net load across the planning horizon for the two extreme cases (no energy storage and 100 kWh of energy storage).

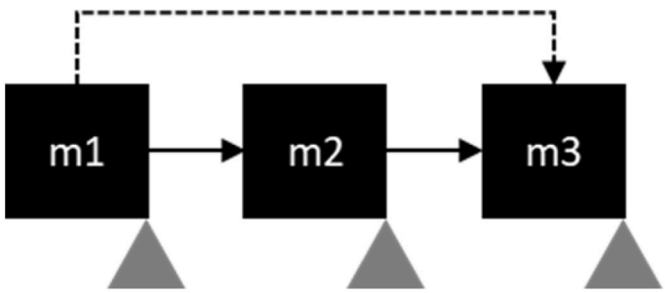


Fig. 1. Illustration of the synthetic manufacturing configuration for exemplary scenario.

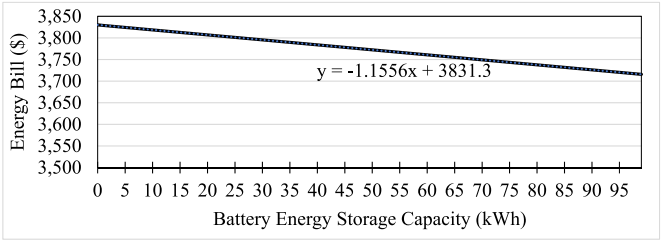


Fig. 2. Optimal energy costs for the billing period with different amount of battery energy storage available for use.

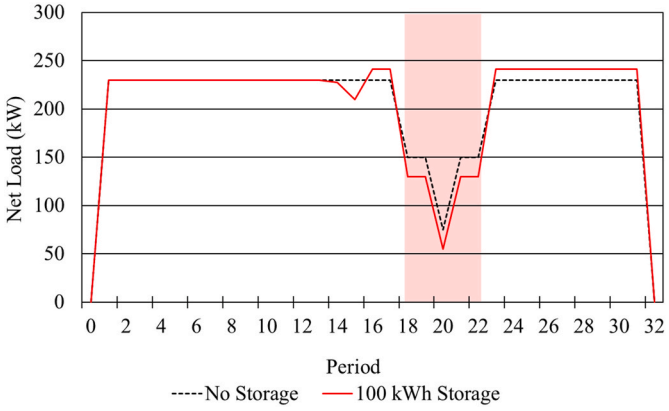


Fig. 3. Net load for two scenarios over the planning horizon (on-peak cost period given by red region).

The slope in Fig. 2 is of particular interest: It tells us that for each unit increase (1 kWh) in available energy storage capacity, the user saves roughly $1.15 on their energy bill. From this we can deduce, that for a given energy storage capacity, the user should only be willing to add capacity if the price is below $1.15 per unit installed. In other words, for this particular user the marginal/shadow price for energy storage is approximately $1.15/kWh. This will be discussed more in the following sections. The reader needs to have in mind that in this paper we consider the investment on energy storage as a cost that has no bearing on the objective function. The addition consideration of initial investment cost is left as future research.

Fig. 3 shows exactly what one should expect: The available energy storage capacity is used during the on-peak cost period in order to avoid the higher on-peak $/kWh rate and reduce the on-peak peak demand. Even in the case of no energy storage, the production of the product is briefly obstructed in order to avoid, in-part, the on-peak cost period. This is accomplished by building a buffer of inventory ahead of decreasing production, which directly highlights the idea of inventory as energy storage.

This simple instance was used to both validate the model and present some interesting insights. In the following section, some more intricate scenarios are introduced, and the results are presented.

4.2. Test scenarios

For the following three scenarios, a day is broken down into 96 15-min intervals. The on-peak period is comprised of intervals 57–76 and the off-peak period is comprised of all others. All on-peak costs are higher than their off-peak counterparts. Taking these rate structures as inputs ties the planning of the manufacturers production directly to the needs of the utility, assuming that the rates are reflective of the utility's goals.

In order to assess the model further, three separate synthetic scenarios were developed with the aim of emulating three different types of generic manufacturing configurations. Scenario 1, depicted in Fig. 4 (a), was designed to emulate a manufacturing operation as a canonical job-shop. Scenario 2, depicted in Fig. 4 (b), was designed to emulate a flow-shop, where product moves sequentially through each machining process. Finally, Scenario 3, depicted in Fig. 4 (c), was designed to be a middle-ground between Scenario 1 and Scenario 2. Specifically, it is comprised of two flow lines with some operations allowed on either of the two lines. In Fig. 4, each of the different arrows represents the possible path of a different product and each box represents a machine/process. Hence, Fig. 4 reveals Scenario 1 has four products, each with many possible paths, Scenario 2 has one product with a single path, and Scenario 3 has two products each with several possible paths. Additionally, each scenario is considered with and without the availability of a battery energy storage device.

As described previously, the goal that the model seeks for each scenario is to move product through the machines/processes such that the demand for each product is satisfied while the energy cost, in addition to the marginal wage increments based on the assignment of workers to various schedules. In addition to being able to shift production around and store energy within inventory buffers, the model can decide how to use the energy storage unit in order to complement the demand shifting. Fig. 5 below shows, for each scenario, the net-load for each period with and without battery energy storage. The shaded region shows the on-peak period. Table 1 shows the time required to find the optimal solution for each scenario and the corresponding optimal value of the objective function. Each scenario was solved using an AMD Ryzen 3900x CPU with 32gb of RAM available.

Aside from a portion of Scenario 2, each scenario was able to schedule production, and use the battery such that on-peak load was eliminated. Note that, for each of the scenarios, there was an increase in load during the end of the day. This is due to Constraint (18) requiring that the energy storage unit be fully charged by the end of the day.

It is important to note here the nature of the results in Table 1. While it would be convenient to compare the scenarios on their value of the optimal objective function, it would be unfair. This is because of the nature of the number of parameters needed to be changed between scenarios in order to accommodate their described operational characteristics. These differences cause a cascade of effects that impact the amount of energy required to produce similar amounts of product. Rather, it is interesting to compare each scenario based on the rigidity, or lack of, in their resultant energy use profile.

The original operation start time for each scenario was period 32. As seen in Fig. 5, for each scenario, the model found that the best solution was to begin as early as possible, even in the face of increased worker wages. Also interesting to note, is that between periods 0 and 45, there is no variation in the load for Scenario 2. This can easily be attributed to the lack of flexibility in a flow-shop style operation, since in a flow-shop there is not much freedom to decide how to move product through the system. On a similar note, the scenario that exhibited the most flexibility in terms of their energy use profile, was Scenario 3. Though, in theory its configuration is less flexible than Scenario's 1 job-shop inspired configuration, here it proved to be advantageous. Finally, the contribution to energy cost savings from the battery energy storage system was ultimately negligible and we likely stand to benefit from a separate process of optimally sizing it to make it more impactful.

Though for these synthetic scenarios, facility adjustments to energy use through production shifting and the use of energy storage did appear to provide some promising results, this may not be the case for other types of operations. That is, other facilities may stand to benefit more or less based on their own unique manufacturing configurations. This opens the conversation for being able to identify those manufacturing operations that are most likely, or most able, to be able to respond to and further some of the utility's goals. Furthermore, we may be able to find groups of facilities, or industries who stand to benefit more because of the energy profiles reflected by their operations. These topics are discussed further in Section 5.

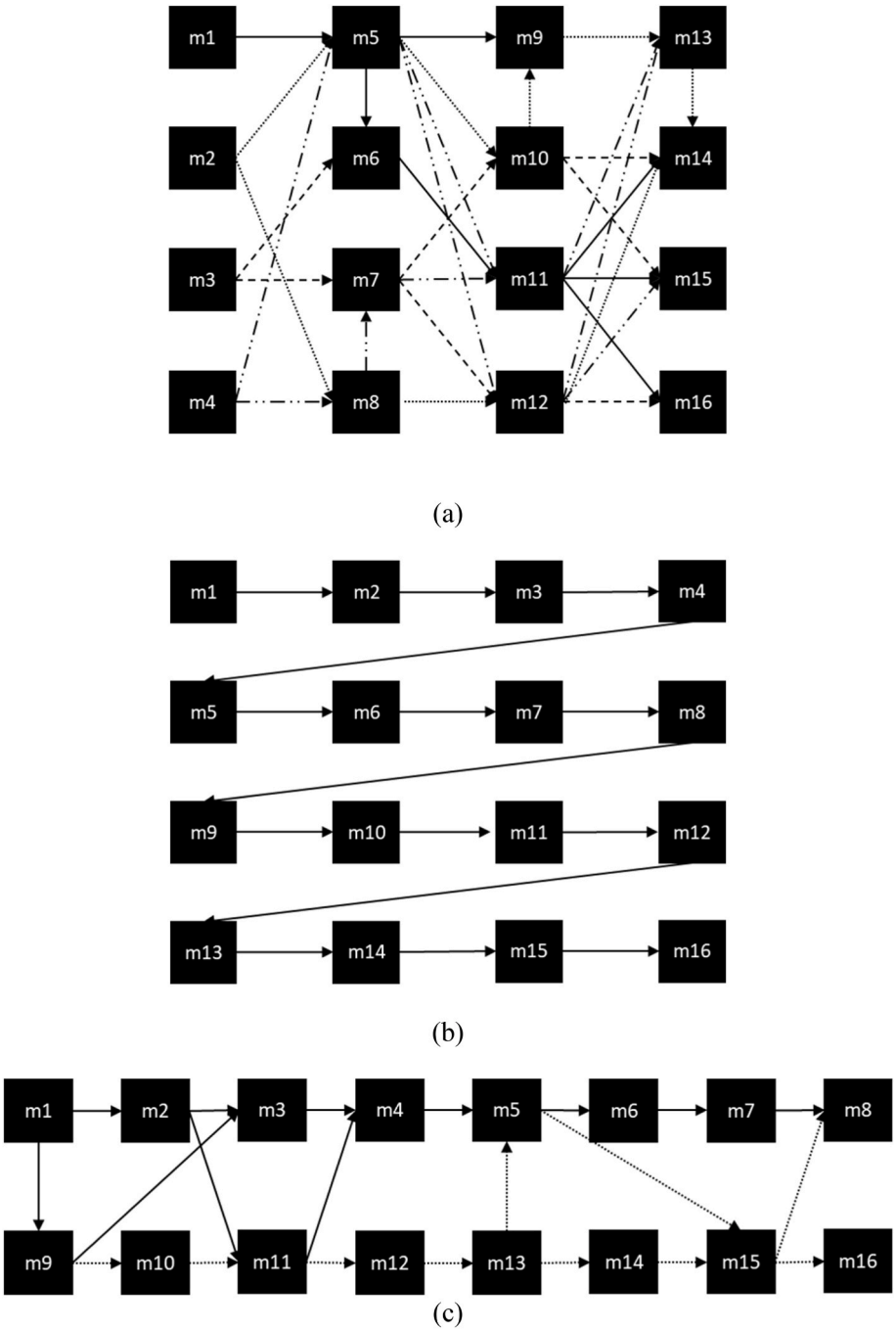
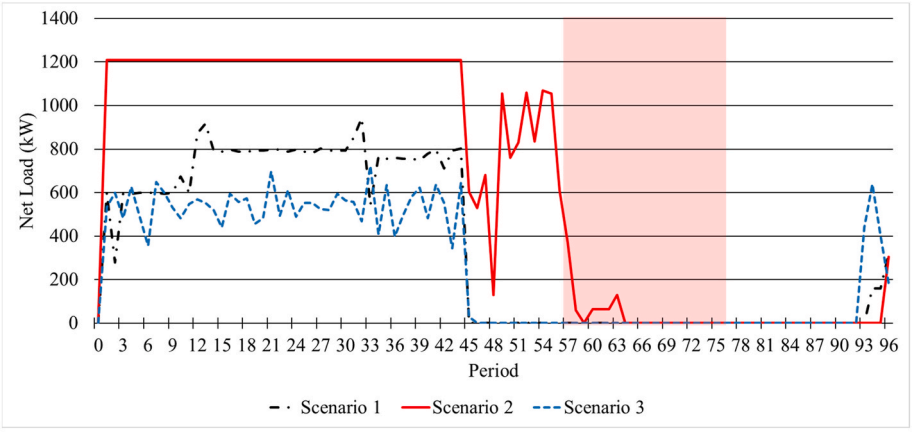


Fig. 4. Illustration of the synthetic manufacturing configurations used as test scenarios.

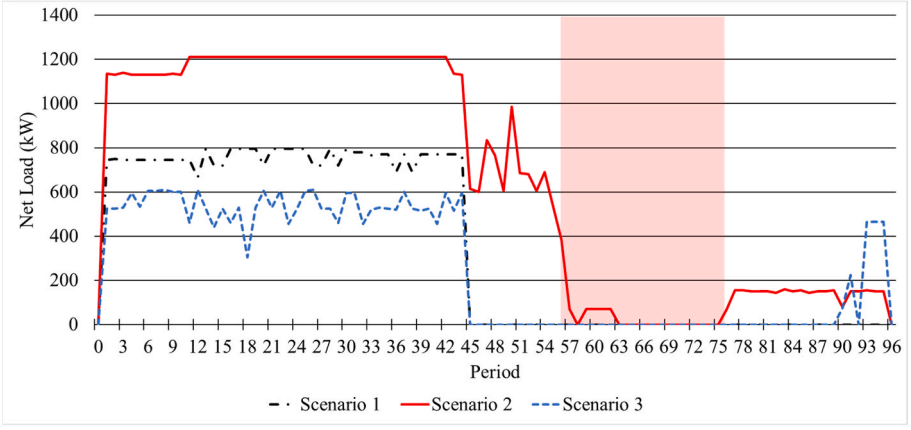
5. Conclusions and future work

In this paper, it was shown that under certain circumstances, manufacturers can be incentivized to use their inventory management capabilities for energy storage. In addition, it was shown that shifting production in parallel with making use of energy storage technology can provide an effective means of capturing value for an individual manufacturer while also aiding the utility to reduce their load during peak demand periods. Most interestingly, there was a clear difference in the adjustment to load profile based on the flexibility of the manufacturing configuration. It was observed that the configuration inspired by a traditional flow shop was least able to adjust its profile through production shifting. This begs the investigation of the potential to classify operations based on their potential to make impactful changes towards their energy use profiles based on the desires of the utility. Hence, an extension of the model presented in the model presented in this work would be as a guide for identifying those candidates whose manufacturing capabilities lend them to participate in DR programs and other ancillary services.

Moreover, it is recommended that individual manufacturers exploit, as much as possible, their ability to capture value through the utility's incentives to adjust the time and magnitude of their energy use. By exploiting this, the users are also directly assisting the utility. Here we use a time-of-use rate structure to take input form the utility. However, more direct communication can be made between the two parties. For example, the utility may wish to send a DR signal to the user detailing the start time, duration, and magnitude of a desired load reduction. The model detailed in this work could then be modified to take these as



(a)



(b)

Fig. 5. Net load for each scenario and period with battery energy storage (a) and without battery energy storage (b).

Table 1
Run time for each scenario.

Scenario	Time to find solution (seconds)	Objective Function Value ($)
Scenario 1	02.74	13,974
Scenario 2	60.71	30,839
Scenario 3	25.64	10,625
Scenario 1 with no battery	01.55	14,242
Scenario 2 with no battery	01.37	31,318
Scenario 3 with no battery	00.86	10,937

parameters and schedule production around this event. This is only one of many ways that further communication can be initiated between the user and the utility.

To summarize, the main contribution offered by this work is the ability for manufacturing to use production scheduling to allow inventory to serve as additional energy storage with all the potential opportunities that this implies. For instance, this new capability would allow the manufacturer to maximize the value captured when taking actions that are incentivized by their utility such as the participation in DR programs. Because this allows both the utility and the manufacturer to gain, it forms the basis for cooperative efforts between the two entities and opens the door for a more dynamic relationship (e.g. manufacturers being able to respond to dynamic pricing signals from the utility in real-time). Hence, an emergent contribution of this paper is the foundation of a negotiation process between the utility and energy consumer(s).

There are a few important areas that are left for future research. Firstly, for each of the scenarios considered, energy storage was assumed to be already available for use. For users who do not already have such capabilities, a new decision variable should be introduced for deciding whether or not to invest in the technology. Considering this would provide some insight on what conditions, both in investment costs and user behaviors, make such an investment attractive. Beyond this, the issue of how to size the system optimally should be considered. Each of these issues will be addressed in future work.

To expand on the idea of using the model results to price the per unit value of energy storage devices, as briefly discussed in Section 3, the model could be adapted to consider different technologies such as solar and other distributed generation technologies. It is proposed that the shadow prices from the model results can be cultivated in order to evaluate which technologies are most promising in terms of economic feasibility amongst different populations of energy consumers. In other words, the model can be used to develop a technology roadmap to most promising technologies for user adoption. This will be the subject of future research resultant from the work presented in this paper. Furthermore, as it has been shown, different manufacturing configurations allow for varying amounts of flexibility when it comes to adjusting their load. If such tools are used to assess this potential, a more strategic impact can be made by next considering clusters of users whose individual contributions, through changes in their energy profile, can be aggregated considering their flexibility to further serve the goals of the utility. If the results of such optimization modeling were shared from manufacturers to the utility, the utility would be able to reap the benefits

of aggregating multiple customers and thus reducing their operational costs while also gaining more insight into customer behavior and incentives regarding their energy use.

The development of the models presented in this paper opens a channel for the utility and the customer to make strategic decisions cooperatively. These models set the stage for a game played between the individual manufacturers and the utility. For example, the developed model will provide insight, for a particular user, into how much energy a user is willing to shift given the prices offered by the utility. With this game played between the utility and many customers, the utility can begin to consider modifications to their existing rate structures until stable solutions are found that benefit both parties. In total, this channel presents the opportunity to develop policies and strategies to aid in making the DSM market more efficient. This investigation is beyond the scope of this work, though the model presented provides the first building block. An additional benefit of the models presented here is the determination of the additional or marginal cost that would make energy storage relevant in reducing the total energy cost of the manufacturing facility. The aggregation of these marginal costs, very often referred to as "shadow prices", help determine the direction that the different storage and generation technologies need to move to become part of the solution in reducing energy costs.

It is also important to note again that the models presented here are completely deterministic and do not capture the stochasticity of the user's load. In the future it will become important to consider these complexities, particularly for those customers with distributed energy resources available to their disposal.

CRediT authorship contribution statement

Miguel A. Peinado-Guerrero: Conceptualization, Methodology, Writing – original draft. **J. Rene Villalobo:** Conceptualization, Methodology, Validation, Funding acquisition, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

Abdel-Hadi, A., Salem, A.R., Abbas, A.I., Qandil, M., Amano, R.S., 2021. Study of energy saving analysis for different industries. J. Energy Resour. Technol. Trans. ASME 143, 1–9. https://doi.org/10.1115/1.4048249.

Alagoz, B.B., Kaygusuz, A., Karabiber, A., 2012. A user-mode distributed energy management architecture for smart grid applications. Energy 44. https://doi.org/10.1016/j.energy.2012.06.051.

Bhattacharya, A., Vasisht, S., Adetola, V., Huang, S., Sharma, H., Vrabie, D.L., 2021. Control co-design of commercial building chiller plant using Bayesian optimization. Energy Build. 246, 111077 https://doi.org/10.1016/j.enbuild.2021.111077.

EIA, 2020. In: Electric Power Annual Industry Report.

Erdinc, O., Tascikaraogolu, A., Paterakis, N.G., Eren, Y., Catalao, J.P.S., 2017. End-user comfort oriented day-ahead planning for responsive residential HVAC demand aggregation considering weather forecasts. IEEE Trans. Smart Grid 8, 362–372. https://doi.org/10.1109/TSG.2016.2556619.

Garey, M.R., Johnson, D.D., S, R., 1976. The Complexity of Flowshop and Jobshop Scheduling Authors (S): M . R . Garey , D . S . Johnson and Ravi Sethi Published by. INFORMS Stable. Accessed : 28-03-2016 13 : 55 UTC Your use of the JSTOR archive indicat. Math. Oper. Res. 1, 117–129. http://www.jstor.org/stable/3689278.

Gelazanskas, L., Gamage, K.A.A., 2014. Demand side management in smart grid: a review and proposals for future direction. Sustain. Cities Soc. https://doi.org/10.1016/j.scs.2013.11.001.

Golmohamadi, H., 2022. Demand-side management in industrial sector: a review of heavy industries. Renew. Sustain. Energy Rev. 156, 111963 https://doi.org/10.1016/j.rser.2021.111963.

GUROBI OPTIMIZATION INC., 2014. Gurobi optimizer reference manual, 2015. URL. http://www.gurobi.com, 29.

Hart, W.E., Laird, C.D., Woodruff, J.W.D.L., Hackebeil, G.A., Nicholson, B.L., Siirola, J. D., Hart, W.E., 2017. Pyomo - Optimization Modeling in Python. Springer. https://doi.org/10.1007/978-3-319-58821-6.

Javadi, M.S., Firuzi, K., Rezanejad, M., Lotfi, M., Gough, M., Catalao, J.P.S., 2019. Optimal sizing and siting of electrical energy storage devices for smart grids considering time-of-use programs. IECON Proc. Ind. Electron. Conf. 4157–4162. https://doi.org/10.1109/IECON.2019.8927263, 2019-October.

Javadi, M.S., Gough, M., Mansouri, S.A., Ahmarinejad, A., Nematbakhsh, E., Santos, S.F., Catalão, J.P.S., 2022. A two-stage joint operation and planning model for sizing and siting of electrical energy storage devices considering demand response programs. Int. J. Electr. Power Energy Syst. 138 https://doi.org/10.1016/j.ijepes.2021.107912.

Lu, X., Li, K., Xu, H., Wang, F., Zhou, Z., Zhang, Y., 2020. Fundamentals and business model for resource aggregator of demand response in electricity markets. Energy 204, 117885. https://doi.org/10.1016/j.energy.2020.117885.

Medha, S., Brenda, C., Mac, K., Trevor, G., 2019. 2019 Utility Demand Response Market Snapshot.

Mouzon, G., Yildirim, M.B., Twomey, J., 2007. Operational methods for minimization of energy consumption of manufacturing equipment. Int. J. Prod. Res. 45, 4247–4271. https://doi.org/10.1080/00207540701450013.

Nagrath, I.J., Kothari, D.P., Desai, R.C., 2008. Modern power system Analysis. IEEE Trans. Syst. Man. Cybern. 12 https://doi.org/10.1109/tsmc.1982.4308783.

Paterakis, N.G., Erdinç, O., Catalão, J.P.S., 2017. An overview of Demand Response: key-elements and international experience. Renew. Sustain. Energy Rev. 69, 871–891. https://doi.org/10.1016/j.rser.2016.11.167.

Peinado-Guerrero, M.A., Campbell, N.A., Villalobos, J.R., Phelan, P.E., 2020. A Framework for Demand-Side Management with Demand Response Input. https://doi.org/10.1115/POWER2020-16635.

Peinado-Guerrero, M.A., Villalobos, J.R., Phelan, P.E., Campbell, N.A., 2021. Stochastic framework for peak demand reduction opportunities with solar energy for manufacturing facilities. J. Clean. Prod. 313, 127891 https://doi.org/10.1016/j.jclepro.2021.127891.

Ribó-Pérez, D., Larrosa-López, L., Pecondón-Tricas, D., Alcázar-Ortega, M., 2021. A critical review of demand response products as resource for ancillary services: international experience and policy recommendations. Energies 14, 846. https://doi.org/10.3390/en14040846.

Rubaiee, S., Yildirim, M.B., 2019. An energy-aware multiobjective ant colony algorithm to minimize total completion time and energy cost on a single-machine preemptive scheduling. Comput. Ind. Eng. 127, 240–252. https://doi.org/10.1016/j.cie.2018.12.020.

Shrouf, F., Ordieres-Meré, J., García-Sánchez, A., Ortega-Mier, M., 2014. Optimizing the production scheduling of a single machine to minimize total energy consumption costs. J. Clean. Prod. 67, 197–207. https://doi.org/10.1016/j.jclepro.2013.12.024.

Tuballa, M.L., Abundo, M.L., 2016. A review of the development of Smart Grid technologies. Renew. Sustain. Energy Rev. https://doi.org/10.1016/j.rser.2016.01.011.

Viana, M.S., Manassero, G., Udaeta, M.E.M., 2018. Analysis of demand response and photovoltaic distributed generation as resources for power utility planning. Appl. Energy 217. https://doi.org/10.1016/j.apenergy.2018.02.153.

Wang, H., Zhong, R.Y., Liu, G., Mu, W.L., Tian, X., Leng, D., 2019. An optimization model for energy-efficient machining for sustainable production. J. Clean. Prod. 232, 1121–1133. https://doi.org/10.1016/j.jclepro.2019.05.271.

Wang, K.J., Dagne, T.B., Lin, C.J., Woldegiorgis, B.H., Nguyen, H.P., 2021. Intelligent control for energy conservation of air conditioning system in manufacturing systems. Energy Rep. 7, 2125–2137. https://doi.org/10.1016/j.egyr.2021.04.010.

Wolsey, Laurence A., 2020. Integer Programming. John Wiley & Sons.